IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 26 August 2024, accepted 21 September 2024, date of publication 30 September 2024,
date of current version 18 October 2024.

*Digital Object Identifier 10.1109/ACCESS.2024.3470970*

RESEARCH ARTICLE

# Effective Edge Solution for Early Detection of Rice Disease on ARM- M Microcontroller

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This work was supported by the Research Consortium of Swinburne University of Technology Sarawak Campus, University of Nottingham Malaysia, Curtin University Malaysia, and Swinburne Innovation Malaysia.

**ABSTRACT** The increasing need for efficient and sustainable agriculture has driven the adoption of advanced technologies like Artificial Intelligence (AI) in crop management. This study evaluates the feasibility of deploying AI models on ARM-M microcontrollers for real-time detection of rice plant diseases, crucial for early intervention and crop health management. We developed and assessed the performance of five Convolutional Neural Network (CNN) architectures: MobileNet V1, MobileNet V2, FD-MobileNet, ResNet, and SqueezeNet. These models were trained on a dataset comprising 5932 high-quality images of rice leaves affected by four prevalent diseases: bacterial blight, blast, brown spot, and tungro. The AI models were optimized and quantized to meet the computational and memory constraints of the ARM-M microcontroller. Our evaluation focused on accuracy, computational efficiency, and resource utilization. The results demonstrated that FD-MobileNet achieves high accuracy of 98.44% with lower computational costs, making it suitable for deployment in resource-constrained environments. The deployment on ARM-M microcontrollers showcased the potential for real-time, on-site disease monitoring, enabling farmers to take timely and effective actions. This study underscores the transformative impact of integrating AI with embedded systems in smart agriculture, promoting sustainable farming practices and enhancing food security.

**INDEX TERMS** Deep learning, ARM-M microcontrollers, edge computing, smart agriculture, resource-efficient AI.

## I. INTRODUCTION

Rice is the primary staple food for a large percentage of the global population, making its production process essential for ensuring food security in many regions [1], [2]. In Malaysia, the agricultural sector faces the challenge of maintaining, if not increasing, domestic rice sufficiency amid various risk factors. Recent statistics from the Department of Statistics Malaysia highlight a concerning trend: the rice self-sufficiency ratio dropped from 65.2% in 2021 to 62.6% in 2022 [3]. This decline underscores the vulnerability of rice production to several factors, such as plant diseases and environmental hazards, particularly in the northern states, which contribute to 60% of the country's rice output [4]. The concentration of rice production in these areas amplifies the agricultural sector's susceptibility to the adverse effects of emerging plant diseases and unfavorable weather conditions.

Early detection of potentially infected plants is crucial to prevent the spread of diseases in plantations, which can incur significant costs in terms of time and financial resources [5]. However, the complexity of procedures for plant disease identification poses challenges for agronomists, especially when dealing with multiple symptoms [6]. Manual disease identification is inefficient and often leads to misdiagnoses and ineffective treatments [7], [8], [9].

The associate editor coordinating the review of this manuscript and approving it for publication was Vishal Srivastava.

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Considering these challenges, there has been a growing interest in leveraging smart agriculture technologies to mitigate risks and enhance productivity. Among these technologies, Artificial Intelligence (AI), particularly Convolutional Neural Networks (CNNs), has shown great potential for transforming plant disease detection and management. CNNs, a type of deep neural network, are well-known for their top performance in image recognition and classification tasks [10], [11]. In the context of agriculture, CNNs have demonstrated exceptional accuracy in identifying various plant diseases through the analysis of leaf images [12], [13]. This capability is critical for early disease detection, which is essential for preventing the further spread of diseases and minimizing crop loss.

Deploying Convolutional Neural Networks (CNNs) poses a significant challenge. Typically deployed on cloud servers, these systems demand stable and uninterrupted internet access, creating difficulties in rural or remote farming areas with limited connectivity. To overcome this, there is a growing need for machine learning solutions that operate directly on edge device, independent of centralized servers and continuous internet access. Edge computing solution, particularly on low-power embedded devices such as ARM-based microcontrollers, offers intelligent, localized solutions that enhance privacy, reduce latency, provide real-time performance, and optimize resource usage

This study explores the deployment of various CNN architectures on ARM-M microcontrollers for real-time, on-site rice plant disease detection. By leveraging the computational power of ARM-M microcontrollers, we aim to provide a practical tool for farmers to detect plant diseases early and take prompt action to mitigate their spread. This approach not only enhances agricultural productivity but also contributes to food security and sustainability.

The findings from this study offer valuable insights into the optimization of AI models for agricultural applications. They highlight the potential of integrating advanced machine learning techniques with portable hardware solutions to address critical challenges in crop management. Through this research, we aim to bridge the gap between laboratory-based disease detection models and their practical application in the field, paving the way for smarter and more resilient agricultural practices.

## II. RELATED WORK

The integration of Artificial Intelligence (AI) with microcontrollers for plant disease detection has gained momentum, leveraging the power of Convolutional Neural Networks (CNNs) on resource-constrained devices for real-time agricultural applications.

A recent report [5] explored the use of dynamic K-means compression to deploy deep neural networks on STM32 microcontrollers for plant disease detection. The study utilized the STM32 microcontroller which provided the necessary computational capabilities while maintaining low power consumption. The compressed neural networks demonstrated high accuracy in detecting various plant diseases, making the solution practical for on-field deployment without compromising performance.

Thai et al. [14] address the challenges of deploying deep learning models for plant disease detection on resource-constrained edge devices. They propose a lightweight hybrid model, Tiny-LeViT, which effectively combines transformer and CNN architectures to achieve high accuracy and faster processing rates on devices like Raspberry Pi and Jetson Nano. Additionally, they introduce a cloud-based collection of 38 public leaf disease datasets, facilitating easier access and preparation for researchers. Their work demonstrates significant improvements in both the efficiency and scalability of AI-driven leaf disease detection in real-world agricultural applications.

Another work [15] introduced a deep learning framework designed to classify rice leaf diseases into nine specific subtypes using a custom VGG16 model. The study utilizes a dataset of 5,932 images, including both self-generated and benchmark datasets, covering various levels of disease severity. The authors propose a customized VGG16 architecture, incorporating additional dropout and dense layers to improve generalization and prevent overfitting. The model achieved a remarkable accuracy of 99.94%, outperforming several state-of-the-art models. The paper emphasizes the importance of early and accurate disease detection in rice plants to ensure sustainable agriculture and reduce the reliance on manual visual inspections.

Joseph et al. addresses the challenge of creating high-quality, real-time datasets for detecting plant diseases in rice, wheat, and maize. The authors developed datasets that include various disease stages under real-world conditions, aiming to enhance the performance of deep learning models for accurate disease classification. Eight pre-trained CNN models, including MobileNet and Xception, were fine-tuned on these datasets, achieving high testing accuracies (above 95%) across different crops. Additionally, a new CNN model, MRW-CNN, was proposed and demonstrated superior performance with testing accuracies of up to 98%. This work contributes to the field by offering robust datasets and models for reliable plant disease detection, essential for sustainable agriculture.

Dubey and Choubey present a robust method for classifying paddy leaf diseases using a hybrid deep learning model that integrates adaptive feature selection techniques. The proposed model follows a four-phase process: preprocessing, feature extraction, feature selection, and classification. Key innovations include the use of Support Vector Machine-Recursive Feature Elimination (SVM-RFE) combined with an Adaptive Rain Optimization Algorithm (ARO) for feature selection, and the application of an Adaptive Bi-directional Long Short-Term Memory (ABi-LSTM) network for classification. This approach significantly enhances accuracy, achieving a classification accuracy of 98.86%, making it highly effective for detecting diseases like Blast, Bacterial Leaf Blight, and Tungro in paddy plant.

Shahidur, Amith, and Tanvir [16] explored the use of IoT-based edge intelligence for rice leaf disease detection. Their system utilized the Raspberry Pi 4 Model B, featuring a quad-core ARM Cortex-A72 CPU. This setup enabled real-time data processing and disease classification using machine learning algorithms. The study achieved high accuracy in identifying rice leaf diseases and demonstrated the feasibility of deploying AI-driven solutions on IoT platforms to enhance agricultural productivity through timely interventions.

Another study [17] conducted a quantitative analysis of a plant disease detector using quantized Convolutional Neural Networks (Q-CNNs) on the STM32 microcontroller. The quantized models achieved 96% accuracy in identifying coffee plant diseases, demonstrating the practicality of deploying compact neural networks on power-restricted devices. This approach provided a scalable solution for real-time, on-site disease detection, essential for improving agricultural practices and ensuring crop health.

Reference [18] introduced a real-time method for corn leaf disease recognition using a deep convolutional neural network (CNN). Their model was optimized and deployed on a Raspberry Pi 3 equipped with the Intel Movidius Neural Compute Stick (NCS), which includes dedicated CNN hardware blocks. The system achieved an accuracy of 88.46% in recognizing corn leaf diseases, demonstrating the feasibility and effectiveness of using such setups for real-time, on-site disease detection without the need for internet connectivity?.

A recent study [19] developed a mobile-based system for maize plant disease detection using deep learning. The researchers evaluated YOLO models (YOLOv3-tiny, YOLOv4, YOLOv5s, YOLOv7s, YOLOv8n) on a dataset of maize leaf images affected by Blight, Sugarcane Mosaic Virus, and Leaf Spot. The YOLOv8n model achieved the highest accuracy of 99.04%. This model was integrated into a mobile app for real-time, on-site disease detection, providing farmers with an efficient tool for early disease identification and management.

The related work section highlights the growing integration of AI, particularly deep learning models, with microcontrollers and edge devices for real-time plant disease detection in agriculture. Various studies have demonstrated the effectiveness of deploying CNNs and other machine learning models on resource-constrained devices like STM32 microcontrollers, Raspberry Pi, and mobile platforms.

While existing research has made significant strides in optimizing deep learning models for resource-constrained environments, there remains a gap in the exploration of more diverse and challenging real-world agricultural conditions. The performance of AI models, particularly deep learning networks, relies heavily on hardware capabilities, which can be limiting in resource-constrained environments. Therefore, it is essential to find a balance between model complexity and computational efficiency to maintain high accuracy while ensuring the feasibility of deployment on ultra-low-power devices.

## III. MOTIVATIONS

To date, deep learning solutions in agriculture have predominantly focused on the development stage, particularly on enhancing the robustness of plant disease detection models at the application layer. While there are existing patents leveraging embedded devices for sunlight adjustment [20] and AI-driven plant disease detection [21], [22], the integration of AI for plant disease detection with embedded devices remains underexplored. Most deep learning models are typically deployed on cloud servers, which necessitates stable and continuous internet access—a significant challenge in rural or remote farming areas [23], [24], [25]. For instance, tea leaf disease detection models placed on Platform-as-a-Service (PaaS) clouds, accessed via smartphones, rely heavily on reliable internet connectivity, which can be a substantial barrier in regions lacking robust internet infrastructure [26].

Traditional deployment of deep learning models often depends on centralized servers with high computational power, which is not always practical or cost-effective for agricultural applications [27]. This approach raises several issues, particularly the high resource requirements associated with centralized servers. Consequently, there is a growing demand for deploying machine learning solutions directly on edge devices, which can function independently of centralized servers and without the need for constant internet connectivity [28]. In many agricultural applications, real-time processing is essential. Ensuring that AI models can process data and produce results in real-time on a microcontroller can be challenging due to the computational and memory constraints.

Edge AI combines Edge Computing and AI, deploying intelligent solutions on microcontrollers close to the data source. This setup offers multiple benefits especially in the context of real-time applications and resource-constrained environments [29], [30]. Microcontrollers enable real-time processing of data, which is crucial for applications like agriculture where immediate responses to detected issues, such as plant diseases, can significantly enhance crop management and productivity by allowing farmers to take immediate action to mitigate the spread of diseases. Furthermore, microcontrollers are typically more cost-effective compared to full-fledged computing systems [31]. This affordability makes advanced AI technologies accessible to a broader range of users, including small-scale farmers and rural communities, thereby democratizing the benefits of AI.

Microcontrollers are designed to operate with low power consumption [32], [33], which is essential for applications in remote or off-grid locations. Deploying AI models on such devices ensures that they can run for extended periods on battery power, making them ideal for field deployments in agriculture and environmental monitoring. The portability and versatility of microcontroller-based systems allow for flexible deployment in various settings, such as handheld devices for in-situ plant health diagnosis.

Edge computing allows data to be processed locally on the device, enabling real-time analysis and decision-making. This is crucial in agriculture where timely actions, such as

detecting and treating plant diseases, can prevent significant crop losses. By processing data locally, edge computing eliminates the need to send data to a central server, which can reduce latency. This is particularly important in remote agricultural areas where internet connectivity may be slow or unreliable.

This approach bridges the gap between laboratory-based disease detection models and their practical application in the field, paving the way for smarter and more resilient agricultural practices.

## IV. HARDWARE

The STM32H747I-DISCO Discovery kit is built around the STM32H747XI microcontroller, featuring an ARM Cortex-M7 and a Cortex-M4 core. This dual-core architecture allows the board to efficiently handle both high-performance and low-power tasks. It also supports X-CUBE-AI, a software extension that enables STM32 microcontrollers to run deep learning models by converting pre-trained neural networks into optimized code. This capability allows the board to perform advanced AI tasks directly on the device, making it ideal for real-time AI processing and edge computing applications without needing external server connectivity.




**FIGURE 1.** STM32H747i-Disco board.



**FIGURE 2.** B-CAMS-OMV camera module.

## V. RICE PLANT DISEASE DATASET DEVELOPMENT

### A. DATA SELECTION

There are many available datasets regarding rice leaf disease. Dataset used in [34] contains 5,932 high-quality images, is specifically curated to focus on four distinct classes of rice leaf diseases. The images are collected on field using NikonDSLR-D5600 with 18–55 mm lens with high resolution.

Another dataset, referenced in [16], comprises 5,447 images across three rice leaf disease classes and one healthy class. However, this dataset was deemed unsuitable due to poor categorization, with some images of infected leaves incorrectly labeled as healthy and vice versa, as noted by the author. Consequently, the primary dataset for this project will be the previous dataset, which offers superior image quality and accurate categorization. Details of this dataset are provided in the table 1.

**TABLE 1.** Details of dataset.

| Leaf Disease | Number of Images |
|---|---|
| Bacterial Blight | 1584 |
| Blast | 1440 |
| Brown spot | 1600 |
| Tungro | 1308 |
| **Total** | 5932 |

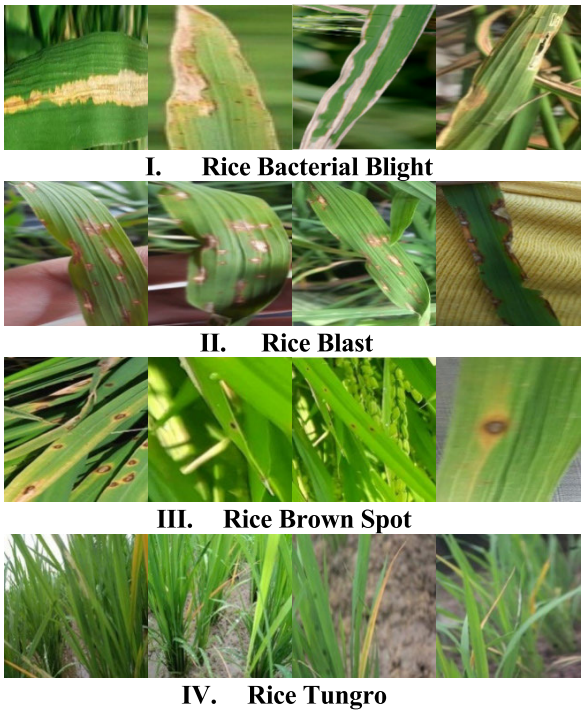



**FIGURE 3.** Samples diseased images from the dataset [35].

### B. DATA PREPROCESSING

The performance of fully convolutional networks is highly dependent on the resolution of input images, as highlighted by various studies [36]. Increasing image size generally improves model performance, but it also increases model

size, which can be problematic for deployment on resource-constrained microcontrollers [37]. Therefore, determining the optimal image size to balance accuracy and deployability requires empirical testing.

For our study, we resized the dataset, employing centre cropping to maintain the important details of the images. This size was chosen to ensure a balance between high accuracy and feasibility of deployment on the ARM-M microcontroller.

Each image was meticulously annotated with the corresponding disease label. To maintain accuracy and consistency, a rigorous cleaning process was implemented to remove any mislabelled or poor-quality images. This step was crucial to ensure that the dataset accurately represented each disease class without any noise.

By carefully resizing, annotating, and cleaning the dataset, we ensured that the input data was of the highest quality, providing a solid foundation for training reliable and accurate neural network models for rice plant disease detection.

### C. DATA AUGMENTATION

The dataset exhibits class imbalance, with varying numbers of images allocated to each disease class. This imbalance can hinder model training, as models trained on imbalanced data tend to be biased towards the majority class. This bias can result in the model having difficulty differentiating the minority class, which is a significant setback for accurate disease detection [38].

Models trained on imbalanced data often show a preference for the majority class due to the abundance of data associated with it. This bias impairs the model's ability to generalize well to unseen data, particularly for the underrepresented classes. To address these issues, data augmentation methods are applied to create a more balanced dataset.

Data augmentation involves transforming existing data to create new, varied samples. Techniques such as image shifting, rotation, zooming, shearing, and flipping are used to increase the dataset's size and diversity. This process ensures more even representation of all classes, reducing the likelihood of the model being biased towards the majority class during training.

For this project, TensorFlow's Keras preprocessing library was chosen due to its suitability for data preprocessing, especially for models built with the TensorFlow Keras API [39]. The ImageDataGenerator within this library is a key tool for generating batches of tensor image data with real-time augmentation. This approach enhances model training and improves generalization by providing a continuous stream of varied training samples.

By employing these data augmentation techniques, the dataset becomes more balanced, allowing the model to learn from a more diverse set of examples. This leads to better performance and accuracy in identifying and classifying all classes of rice plant diseases, ultimately contributing to more reliable and robust disease detection.

## VI. MODEL DEVELOPMENT

Since the STM32 platform only supports 8-bit quantized models, the choice of deployable models is limited [40]. Five state-of-the-art models were selected for deployment on the STM32H747I-DISCO board, each offering unique advantages for different applications. These models are FD-MobileNet, MobileNet V1, MobileNet V2, ResNet V1, and SqueezeNet.

### A. MobileNet V1

MobileNet architecture gives an efficient approach for computer vision applications on mobile and embedded devices. It uses depthwise separable convolutions with the main intention of reducing computational cost and model size without altering much accuracy [41]. Depthwise separable convolution is a 2 stages convolution starts with a depthwise convolution followed by pointwise convolution [42].

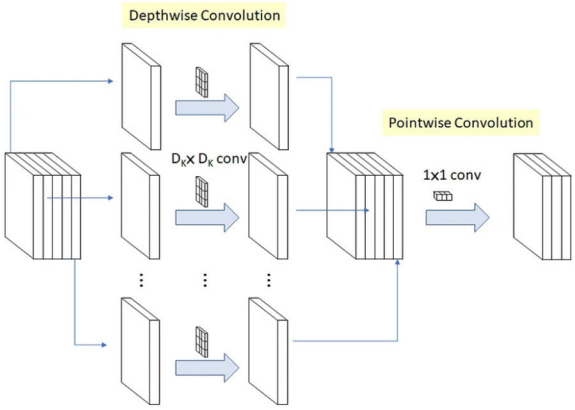



**FIGURE 4.** Depthwise separable convolution [43].

MobileNets V1 demonstrate impressive performance on a range of different tasks like image classification, object detection, and face recognition, consequently, showcasing their versatility. The architecture's adaptability makes it especially appropriate for applications with limited computational resources, by giving the best possible balance between efficiency and performance. MobileNets V1 have emerged as an extraordinary deep learning model that can now be employed in edge devices. This represents a major step forward in the democratization of AI technologies and their practicality to real-world applications.

### B. MobileNet V2

MobileNet V2 introduces significant improvements over MobileNetV1, most notably through the introduction of the inverted residual structure with linear bottlenecks. The figure below shows the difference between MobileNet V1 and MobileNet V2.

There is slight change in the structure of convolution layer. The architecture begins with a low-dimensional compressed representation through a pointwise convolution then expands it to higher dimensions through depthwise convolution and

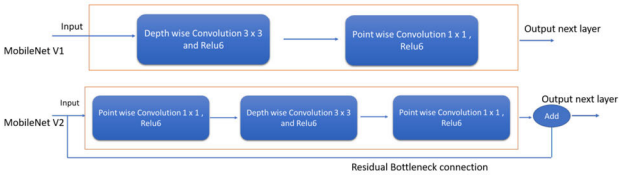



**FIGURE 5.** Difference between MobileNet V1 and MobileNet V2 [44].

then projects it back to a low dimension with a pointwise convolution. ReLU 6 is not mentioned in their MobileNet V1 paper, however, in MobileNet V2, it has mentioned that ReLU 6 is used for comparison in both V1 and V2 versions. ReLU 6 is used due to its robustness when used in low-precision computation [45]. The second point wise convolution in MobileNet V2 is without any non-linearlity so that the deep networks will not only have the power of a linear classifier on the non-zero volume part of the output domain [44].

### C. FAST-DOWNSAMPLING MobileNet (FD-MobileNet)

Fast-Downsampling MobileNet (FD-MobileNet), inspired from Mobilenet, provides a good accuracy on various image classification tasks for very limited computational budgets. This faster down-sampling is achieved by consecutively applying depthwise separable convolutions with larger strides at the beginning of the network, without the use of max pooling to avoid additional computational overhead.

In a slow downsampling strategy, the downsampling is performed in the later layers of the network instead, and therefore, the layers have larger spatial scales. Conversely, fast downsampling strategy perform downsampling at the beginning of the network, which significantly reduces the computational cost. The faster downsampling is achieved by consecutively applying depthwise separable convolutions with large strides at the beginning of the network. The max pooling is not introduced as it does not help in improve accuracy but wasting extra computation [46].

### D. ResNet V1

As the AlexNet came into the picture in 2012, developers found that developing deeper Convolutional Neural Networks (CNNs) could highly boost the accuracy of the models by extracting the subtlest of features from the image data [47]. However, this strategy encountered a major hurdle: the vanishing gradient problem had led to the limitation in achieving further enhancements in the depth and complexity of neural networks. In machine learning, in the process of network training with gradient descent and backpropagation, the amount of each weight to adjust is based on the error function's derivative relative to that weight. While the gradient becomes noticeably smaller, this makes the adjustments to the weights less profound and the accuracy becomes stagnant, or even plateau [48]. This phenomenon, called the vanishing gradient problem, thus, makes the network's learning process more difficult.

ResNet distinguishes itself from traditional neural networks by focusing on learning the residual, or difference, between input and output within its blocks, rather than directly approximating the output function.

$$F(x) = H(x)x$$
$$H(x) \ : true\ output\ of\ the\ residual\ block | x : the\ input \quad (1)$$

This design strategy shifts the learning objective from directly approximating $H(x)$ to approximating $F(x)$, the difference between the input and the true output. This allows ResNet to set $F(x) = 0$, meaning the output of the layer is same as its input. This is powerful as it simplifies the learning process instead of forcing every layer to calculate a complex function, it allows some of the layers to directly send the input to next layer. This shortcut mechanism eliminates the vanishing gradient problem by ensuring that there is a propagation of learning signal through various networks without weakening, hence, making the interpretation of deeper networks easier.

### E. SqueezeNet

SqueezeNet utilizes a unique "squeeze" and "expand" strategy in its fire modules to reduce the number of parameters while maintaining high performance [49]. The key innovation of SqueezeNet lies in its fire modules, which consist of a squeeze layer that uses $1 \times 1$ convolutions to reduce the number of input channels, followed by an expand layer that applies a mix of $1 \times 1$ and $3 \times 3$ convolutions [50]. This architecture enables SqueezeNet to achieve AlexNet-level accuracy with 50x fewer parameters, making it highly suitable for deployment on resource-constrained devices, such as microcontrollers and mobile platforms.

SqueezeNet's compact size and efficient performance make it an ideal choice for applications requiring real-time processing and low power consumption, without compromising on accuracy.

## VII. MODEL TRAINING AND QUANTIZATION

The collected, pre-processed, and augmented datasets are split 80% for training and 20% for testing and validation [51]. This split ensures that the model has a substantial amount of data to learn from while also providing a separate dataset for evaluating its performance and generalization capabilities. The training set is used to train the models, enabling them to learn the patterns and features associated with different rice plant diseases. The testing and validation sets are crucial for assessing the models' performance and tuning hyperparameters to avoid overfitting.

Once the datasets were split, the training process began by feeding the training data into the selected CNN models: MobileNet V1, MobileNet V2, FD-MobileNet, ResNet, and SqueezeNet. Each model underwent rigorous training, where the parameters were kept constant throughout the experiment. Data augmentation techniques such as rotation, flipping, and scaling were applied during training to enhance the model's

robustness and improve its ability to generalize to unseen data.

After training, the models were quantized to reduce their size and computational requirements, making them suitable for deployment on the STM32 board. The quantization process involved converting the model weights to lower precision formats 8-bit integers TensorFlow Lite format, which significantly reduced the memory footprint and inference time without compromising much on accuracy.

## VIII. DEPLOYMENT

The training and deployment of a TensorFlow model onto an STM32 board, as depicted above, culminates with the model in a TensorFlow Lite file format. To operationalize this on the STM32 hardware, it is necessary to convert the TensorFlow Lite file into a .cc source file. This can be done through the STM32Cube.AI Developer Cloud, which offers an online platform and services for creating, optimizing, benchmarking, and generating AI models tailored for STM32 microcontrollers. This cloud-based solution provides a more user-friendly, visual approach to converting and managing AI models for STM32, making it the preferred choice for this project.

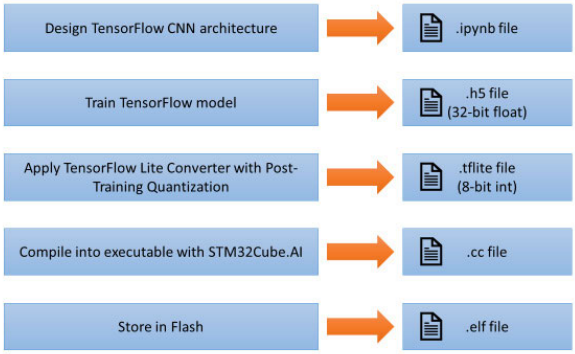



**FIGURE 6.** Method to deploy model on STM32 board [17].

The conversion of the AI model to the STM32 microcontroller environment was marked by the generation of a suite of C files, each serving a distinct role in the deployment process. To streamline the integration of these components, the FP-AI-Vision and Model Zoo package was employed. This comprehensive package supplies a pre-configured STM32CubeIDE project with all the necessary libraries and modules to support computer vision applications. It includes the Hardware Abstraction Layer (HAL) library, Board Support Package (BSP) component libraries for the LCD display and camera module, image processing libraries, and the baseline configuration settings for the STM32 board. The inclusion of the AI model's C source and header files into this project's workspace effectively integrates the software with the board's hardware.

The diagram details the data flow and processing steps within the embedded system, highlighting the interaction between hardware and software components. The process

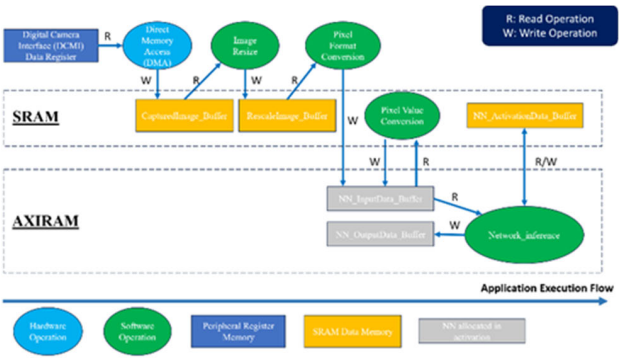



**FIGURE 7.** Memory layout.

begins with the Digital Camera Interface (DCMI) Data Register reading image data, which is then transferred via Direct Memory Access (DMA) to CapturedImage_Buffer. Next, the image undergoes resizing, and the resized image is stored in RescaledImage_Buffer. Following this, there's a pixel format conversion involves reading from the RescaledImage_Buffer, converting the image format to RGB888, and writing back to another buffer. Subsequently, pixel value conversion occurs, formatting the pixel values to uint8 compatible with the neural network. The data is then ready for the NN_InputData_Buffer, located in SRAM, where the image is stored for processing by the neural network. Network inference is then performed, which is a read/write operation involving both the NN_InputData_Buffer and an NN_ActivationData_Buffer, related to the model's layers and neurons' activations. The inference result, or NN_OutputData_Buffer, is written to memory as part of this process. SRAM and AXI RAM (AXIRAM) are shown as peripheral data memory and is part of the system memory hierarchy where data is stored and manipulated during the network inference process. The inference results are moved to SRAM as part of the post-inference process. Overall, the diagram indicates a sequence of operations and data transformations from image capture to preparing the input for neural network inference, culminating in the generation of an output that could be used for further applications or display. The steps of our deployment step can be found here (https://github.com/jingxan/Real-Time-Rice-Plant-Disease-Detection-with-ARM-M-Microcontrollers).

Once the application code was complete, the project was compiled and flashed onto the STM32 board. The real-time inference capabilities were tested by capturing images of rice leaves through the connected camera module. The system processed these images using the deployed AI models, and the results were displayed on the onboard LCD screen.

## IX. RESULT

### A. EXPERIMENT SETUP

In this study, five state-of-the-art lightweight Convolutional Neural Network (CNN) models were selected for training on a meticulously processed dataset of rice leaf images. The software implementation was carried out using TensorFlow,

Keras, and CUDNN, which provided the necessary tools and libraries for deep learning model development and optimization. The models were trained on an NVIDIA GeForce RTX 3060 GPU, leveraging its high computational power to accelerate the training process and handle the extensive data augmentations applied to the dataset.

After training, the CNN models were deployed on the STM32 board, a versatile platform featuring the STM32H747XI microcontroller. This microcontroller, with its dual-core ARM Cortex-M7 and Cortex-M4 architecture, is well-suited for handling the computational demands of real-time neural network inference. The board was connected to a B-CAMS-OMV camera module bundle, providing the necessary hardware to capture high-resolution images of rice leaves in real-time.

To facilitate portability and ease of use in the field, a handheld device was designed to house both the STM32 microcontroller and the camera module. This device was powered by a 10,000mAh power bank, ensuring extended operation time and reliability during field tests. The design considerations also included ensuring that the device was lightweight and ergonomic, making it suitable for prolonged use by farmers and agricultural technicians.

The accuracy of the trained models was first evaluated through simulations on a laptop, where the test images from the dataset were fed into the models to assess their performance metrics, including precision, recall, and F1 score. This step was crucial to ensure that the models were performing as expected before being deployed on the hardware.

Following the simulation tests, five volunteers were recruited to participate in real-world testing of the device. These volunteers were provided with a handheld device and tasked with capturing images of rice leaves under various conditions. The primary objective was to evaluate the models' performance in detecting and classifying rice leaf diseases in a practical setting. The results from these tests were compared with the simulation results to assess the consistency and reliability of the models when deployed on the STM32 microcontroller.

This comprehensive experiment setup ensured that the models were thoroughly tested and validated, both in controlled simulations and real-world scenarios, providing valuable insights into their efficacy and robustness for practical agricultural applications.

### B. HYPERPARAMETER SETUP

To ensure a fair comparison among the five CNN models, identical training hyperparameters were used. All models were optimized using the ADAM optimizer with a learning rate of 0.001. A batch size of 64 was selected for the training process. The dropout rate was set to 0.5, and each model was trained for 50 epochs. Additionally, a learning rate scheduler, set to reduce on plateau, was employed to adjust the learning rate dynamically based on the training performance.

### C. EVALUATION INDICES

In evaluating the performance of the CNN models, a confusion matrix was employed to provide a detailed breakdown of the classification results [52]. A confusion matrix is a square matrix used to visualize the performance of a classification model by comparing the predicted labels against the actual labels. It includes four critical metrics:

- **True Positives (TP)**: Instances where the model correctly identifies diseased leaves.
- **True Negatives (TN)**: Instances where the model correctly identifies healthy leaves.
- **False Positives (FP)**: Instances where the model incorrectly classifies healthy leaves as diseased.
- **False Negatives (FN)**: Instances where the model fails to identify diseased leaves.

The confusion matrix allows for a comprehensive assessment of the model's accuracy, precision, recall, and F1 score. Accuracy measures the proportion of true results (both true positives and true negatives) among the total number of cases examined [53].

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ Number\ of\ Predictions} \tag{2}$$

Precision is the ratio of correctly predicted positive observations to the total predicted positives. It is calculated as:

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Recall measures the ability to identify all actual positive instances.

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

The F1 score provides a balance between precision and recall, making it a valuable metric for evaluating the overall performance of the classification models.

$$F1Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

In this experiment, the confusion matrix was instrumental in highlighting the models' strengths and areas for improvement. The calculated metrics allowed for a thorough evaluation of each model's performance, identifying how well they could distinguish between healthy and diseased leaves.

### D. PERFORMANCE COMPARISON OF CNN MODELS

To evaluate the computer resources consumed by a model, an essential metric to consider is Multiply-Accumulate Operations (MACCs). A model with a high number of MACCs requires more operations to process data, indicating greater hardware resource usage. These operations are prevalent in data processing and machine learning but can strain hardware resources [54]. Flash weight optimization plays a crucial role in managing memory usage during model inference by efficiently encoding model weights. A model with excessive flash weight demands may encounter difficulties in memory

**TABLE 2.** Results of evaluation of 5 state-of-the-art model.

| Model Type | MACCs (M) | Ram Activations (KiB) | Flash Weights (KiB) | Inference Time (ms) | Quantized Model Accuracy, % |
|---|---|---|---|---|---|
| Fd_mobilenet | 3.962872 | 67.46 | 197.15 | 19.81 | 98.75 |
| Mobilenet v2 | 19.10 | 262.22 | 543.41 | 100.91 | 98.38 |
| Resnet v1 | 200.64 | 546.05 | 138.19 | 580.98 | 96.78 |
| SqueezeNet v1 | 81.21 | 262.18 | 803.34 | 261.25 | 99.22 |
| Mobilenet v1 | 13.612792 | 88.87 | 300.44 | 53.93 | 97.47 |

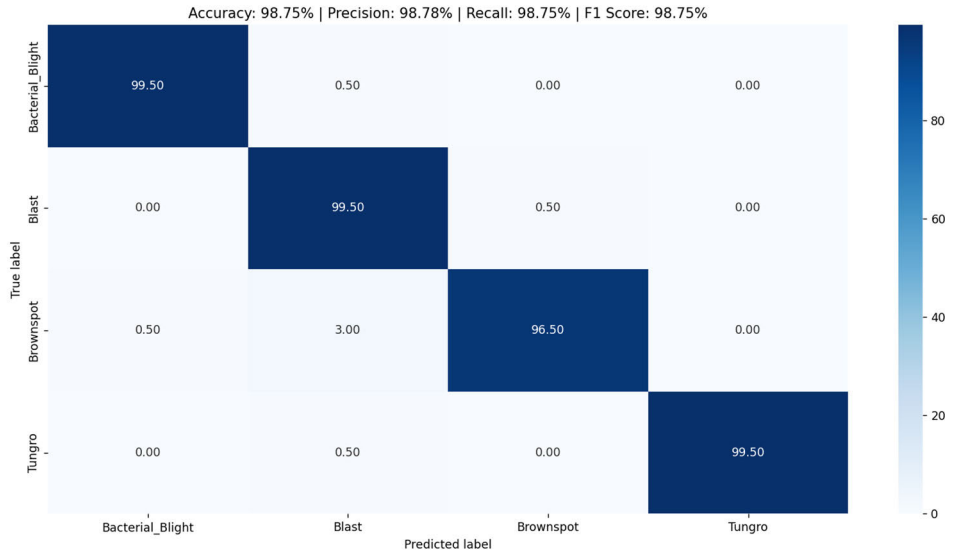



**FIGURE 8.** FdMobileNet confusion matrix.

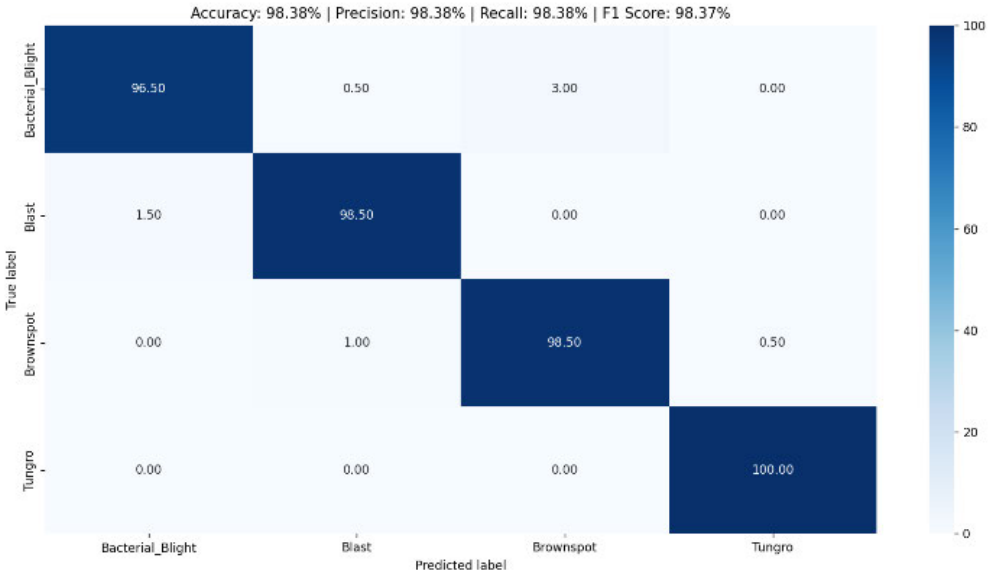



**FIGURE 9.** MobileNet V2 confusion matrix.

management [55]. Additionally, RAM Activation refers to the utilization of Random Access Memory (RAM) in a computing system. Models which require substantial RAM usage can put a strain on system memory resources [56]. Inference time, on the other hand, gauges the speed at which a model generates predictions or responses after receiving input. Ideally,

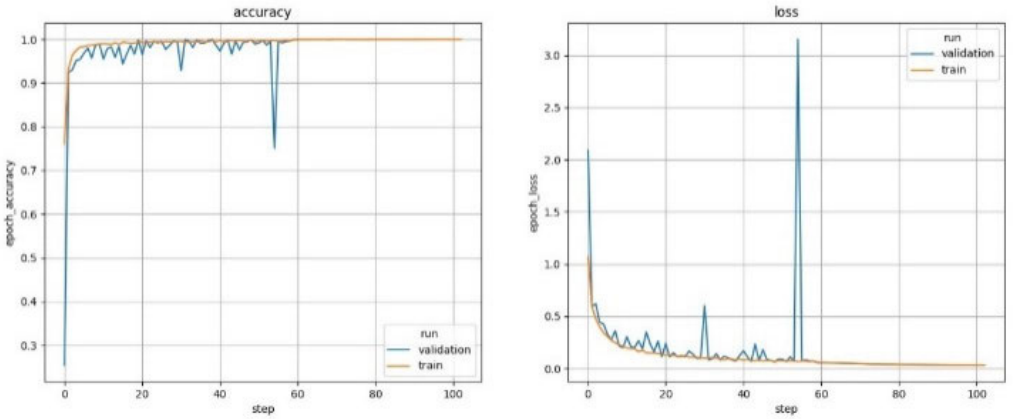



**FIGURE 10.** FdMobileNet training curve.

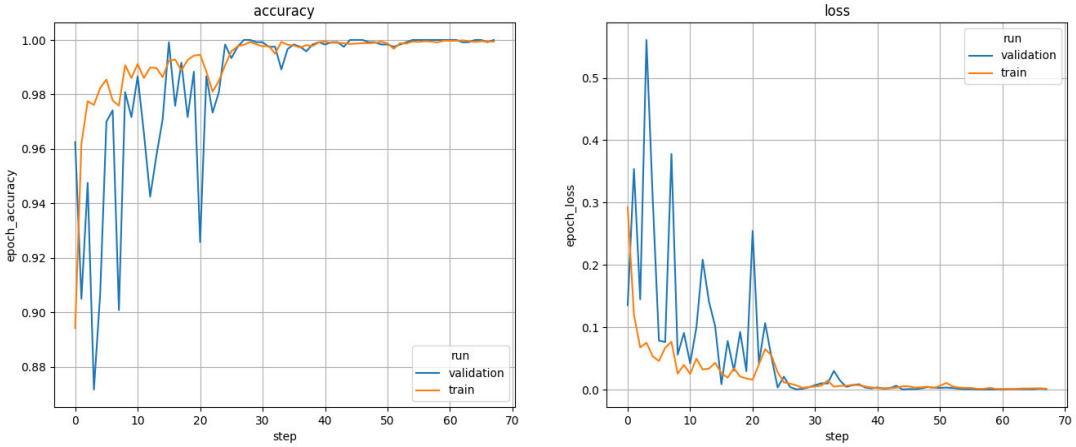



**FIGURE 11.** MobileNet v2 training curve.

a model should strive for the shortest possible inference time to enhance efficiency and responsiveness. Table 2 shows the results recorded from the experiment.

The results reveal that FdMobileNet is the most resource-efficient model, requiring the least number of MACCs, and achieving the shortest inference time while still delivering a high accuracy rate of 98.44%. Although SqueezeNet achieves the highest accuracy, its large model size poses deployment challenges on the STM32 board. MobileNet V2 ranks second in terms of accuracy and demonstrates low computational resource consumption.

### *E. EVALUATING MODEL PERFORMANCE ON MICROCONTROLLER*

MobileNet V2 and FD-MobileNet models were selected for deployment on the microcontroller due to their lightweight architecture and high accuracy. The experiment was conducted to simulate real-world conditions while maintaining a controlled environment. This simulation involved displaying test images on a screen, using a test file comprising 30 randomly selected images for each class drawn from the test image files used in simulation testing.

For practical applications, a handheld device was designed using SOLIDWORKS and 3D printed for physical deployment. This handheld device houses the STM32 microcontroller and camera module, facilitating real-time rice plant disease detection in a portable and user-friendly format. The ergonomic design ensures ease of use for farmers and agricultural technicians, allowing them to capture images of rice leaves and receive immediate diagnostic feedback on the device's screen. This setup simulates the field environment, providing a realistic assessment of the models' performance and their potential impact on agricultural practices.

To closely mimic the conditions that would be faced in a paddy field, the camera was not fixed in terms of distance or angle during the evaluation process. This setup is representative of the practical challenges encountered when the device is used handheld in the field, where maintaining consistent camera distances or angles while capturing images can be difficult. To assess the model's accuracy under these variable conditions, five volunteers were recruited to operate the device and collect testing data. The average accuracy from these trials was subsequently calculated, providing a reliable indicator of the model's performance under realistic usage scenarios.

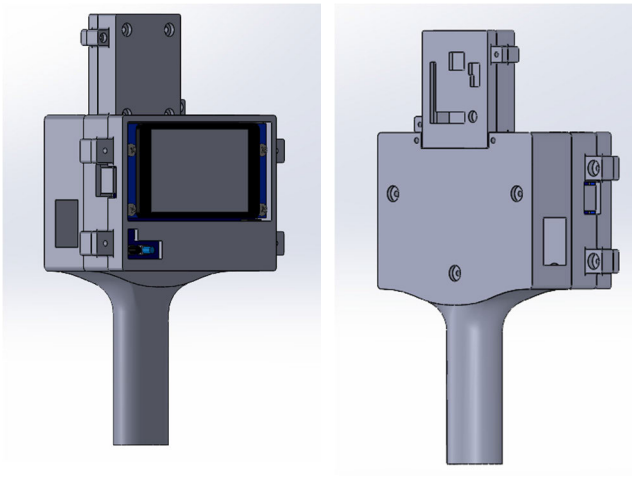


**FIGURE 12.** CAD design for handheld application.

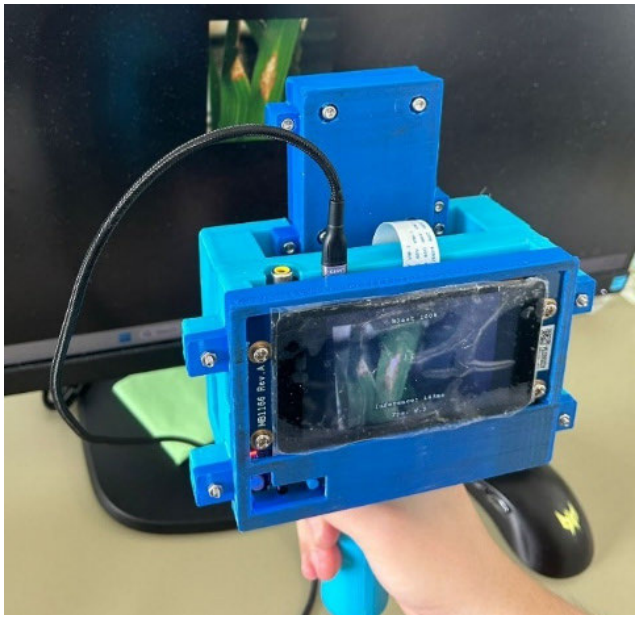


**FIGURE 13.** Evaluating model performance on hardware.

As shown in Table 3, the model achieved a 90% accuracy rate across 120 images. This performance, while commendable, is below the simulation accuracy of 98.75%. Several factors inherent to the real-world deployment environment, such as lighting conditions, camera distance, and angle variability, contributed to this discrepancy. Unlike in simulations where test images are directly inputted into the model, the hardware deployment introduces variables like screen brightness and ambient light, which can affect image capture. Furthermore, the variability in user handling—given that different users operated the device without specific instructions on zooming or angling—reflects the diverse conditions under which the device is expected to perform in practical settings.

Notably, the model showed high effectiveness in recognizing bacterial blight and blast, with accuracies nearing 97%, suggesting that these conditions are well-captured by the model despite environmental variables. However, the performance was notably lower for brown spot and tungro diseases, with a significant number of false positives, potentially indicating sensitivity to the environmental factors.

**TABLE 3.** Validation result of FD_MobileNet model.

| Class | Number of False Positive |
|---|---|
| Bacterial Blight | 1 |
| Blast | 1 |
| Brown Spot | 3 |
| Tungro | 7 |
| **Total** | **12** |

**TABLE 4.** Validation result of MobileNet V2 model.

| Class | Number of False Positive |
|---|---|
| Bacterial Blight | 0 |
| Blast | 0 |
| Brown Spot | 1 |
| Tungro | 2 |
| **Total** | **3** |

Table 4 presents the performance analysis of MobileNet V2, which exhibits superior accuracy at 97.5%. It is shown that MobileNet V2 achieves perfect identification of bacterial blight and blast diseases, outperforming FD-MobileNet in these categories. This highlights MobileNet V2's effectiveness and reliability in accurately diagnosing these specific rice plant diseases. Its advantage over FD_MobileNet is notably evident in the detection of Tungro, where MobileNet V2 achieves 93% accuracy, significantly higher than FD_MobileNet's 77%. This discrepancy underscores the architectural differences between the two models. MobileNet V2's complex structure excels in identifying subtle features, although it demands more computational resources. Conversely, FD_MobileNet utilizes a fast-downsampling strategy, rapidly reducing spatial dimensions by 32× within just the first 12 layers—half the layers utilized by its predecessor, MobileNet. This efficient approach enables FD_MobileNet to effectively recognize diseases with distinct visual markers, such as the white areas indicative of bacterial blight, the white spots characterizing blast, and the brown spots defining brown spot disease. However, it struggles with Tungro detection, a disease lacking distinctive visual cues in its training images, rendering it challenging to distinguish from other conditions based solely on green leaf appearance. This challenge in identifying Tungro with FD_MobileNet, but not in simulation results, highlights the impact of environmental factors in hardware evaluations. In simulations, direct image feeding to the model ensures clarity and accuracy, whereas, in hardware tests, external variables introduce complexities not present in controlled environments.

## X. CONCLUSION AND FUTURE WORK

### A. CONCLUSION

This study successfully demonstrated the feasibility and practicality of deploying lightweight Convolutional Neural Network (CNN) models on ARM-M microcontrollers for real-time rice plant disease detection. Through a comprehensive evaluation, MobileNet V2 and FD-MobileNet emerged as the most suitable models due to their high accuracy and efficiency. These models were rigorously trained using a meticulously processed dataset of rice leaf images, optimized for deployment, and tested both in simulations and real-world scenarios.

The deployment process involved several critical steps, including the conversion of trained models into a format compatible with the STM32 board using the STM32Cube.AI Developer Cloud, and the integration of these models into a handheld device designed for practical field use. The handheld device, equipped with a camera module and powered by a portable power bank, provided real-time diagnostic feedback to users, enabling effective disease management in rice agriculture.

The real-world testing phase, although limited to simulated conditions, showed promising results. The models were able to accurately classify rice leaf diseases, providing a reliable tool for farmers to detect and manage plant diseases promptly. This application has the potential to significantly reduce crop losses and improve yield by enabling early detection and intervention.

Moreover, this study highlights the transformative impact of integrating advanced AI techniques with embedded systems in agriculture. By leveraging the computational power of ARM-M microcontrollers, we were able to create a practical and efficient solution for real-time plant disease detection. This approach not only enhances agricultural productivity but also contributes to sustainable farming practices and food security.

### B. FUTURE WORK

Future research will focus on optimizing the CNN models to further enhance their performance and efficiency for deployment on ARM-M microcontrollers. Model optimization will involve fine-tuning hyperparameters such as learning rate, batch size, dropout rate, and the number of epochs to achieve the best possible balance between accuracy and computational efficiency. Selecting an appropriate learning rate is essential for efficient model training. A high learning rate may cause the model to converge prematurely to a suboptimal solution, whereas a low learning rate can result in prolonged training times or entrapment in local minima. The batch size, which denotes the number of training samples used to update the model's weights in each iteration, is also critical. Larger batch sizes can enhance the stability of the training process but require more memory, while smaller batch sizes allow the model to be more responsive to changes in the training data, albeit with increased noise in the gradient updates. Additionally, the dropout rate, employed to prevent overfitting by randomly omitting units during training, must be carefully calibrated to ensure the model generalizes effectively to new data without compromising performance. Hyperparameter tuning can be performed using techniques such as grid search, random search, or more advanced methods like Bayesian optimization, which systematically explore the hyperparameter space to identify the optimal configuration.

In addition to model optimization, expanding the dataset is a critical component of future work. The current dataset, comprising 5,932 high-quality images of rice leaves affected by four diseases, has provided a solid foundation. The dataset will be expended, and the expanded dataset will include images captured from various geographic locations and under different environmental conditions. By incorporating this diversity, the model will be better equipped to identify and classify diseases accurately in real-world scenarios, ultimately enhancing its performance and reliability in practical agricultural applications. Various deployment strategies will also be explored to optimize the balance between parameter efficiency and memory allocation during model deployment. This will involve investigating techniques that allow for the fine-tuning of model parameters to maximize performance while minimizing the memory footprint on resource-constrained devices. This exploration will provide insights into achieving an optimal balance that supports the practical application of advanced AI solutions in real-time agricultural scenarios.

By expanding the dataset and optimizing the models, future research aims to develop a more effective and scalable solution for real-time rice plant disease detection. This will ultimately contribute to more sustainable agricultural practices, enhanced crop management, and improved food security.

Future work will focus on extensive field testing of the models using the handheld device in diverse agricultural settings. This step is essential to validate the models' performance under real-world conditions and ensure their reliability and effectiveness for practical use.

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